

Bridging Few-Shot Meta-Learning and Strong Tabular Baselines for ICU Treatment Feasibility Prediction on MIMIC-IV

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Abstract

Background. Few-shot meta-learning achieves strong results in computer vision and NLP when pretrained representations exist. *Tabular* clinical prediction from moderate-sized intensive care unit (ICU) cohorts is still dominated by tree ensembles and regularized linear models, raising the question of whether metric-based meta-learning can be made competitive without misleading claims of “parity.” **Objective.** To evaluate a complete pipeline—Symptom Tokens, transformer encoder (ETHOS), prototypical networks, advanced training (contrastive pre-training, knowledge distillation, adaptation), and neural-classical stacking—against strong tabular baselines on a reproducible treatment feasibility task from MIMIC-IV v3.1.

Methods. We extracted $n=2,000$ first ICU stays (five disease strata), 25-D tabular features from 15 labs and 7 vitals in the first 24 h plus engineered summaries, and fixed 64%/16%/20% train/validation/test splits (seed 42). Experiments included (1) classical baselines and tuned gradient boosting, (2) episodic K -shot sensitivity, (3) federated vs. centralized few-shot training, (4) component ablations, (5) cross-node generalization. We report a clear *reporting hierarchy*: tabular ceiling $\rightarrow$ stacked hybrid $\rightarrow$ few-shot-only $\rightarrow$ federated sensitivity.

Results. Repository-frozen exports give: tabular strong model **AUROC 0.747** (augmented RF variant, `exp_final_best_model.csv`); tuned RF in the tabular-SOTA grid **0.744**; stacked BEST few-shot+classical **0.746** (95% bootstrap CI 0.699–0.797); few-shot-only BEST (GFA+CrossAttn) **0.670** (CI 0.612–0.726). The main script comparison (`exp1_main.py`) yields **Proposed_FewShot 0.611** vs. Random Forest **0.762** on the same split—showing that episodic baselines and fully engineered stacks are not interchangeable. Federated simulation reduces AUROC by ≈ 0.005 vs. centralized few-shot training.

Conclusions. With disciplined reporting, few-shot pipelines can approach tabular ceilings when classical predictions and embeddings are fused via calibrated stacking. Claims of non-inferiority require paired tests on aligned score vectors. Limitations include single-center data and simplified labels.

Keywords: few-shot learning; electronic health records; MIMIC-IV; prototypical networks; stacking; knowledge distillation; ICU; reproducibility

1 Introduction

1.1 Clinical motivation

ICU capacity, readmission penalties, and bundled payment models make *early* prediction of whether a patient is on a trajectory toward a safe, timely discharge clinically and economically relevant. Machine learning on structured labs and vitals supports such decision support [1, 5, 15], but hospital data are often moderate in scale (hundreds to a few thousand labeled stays per site), heterogeneous across institutions, and subject to strict privacy rules. These constraints motivate both **strong tabular baselines** and **data-efficient** neural approaches.

1.2 Few-shot learning in tabular clinical settings

Few-shot classification aims to generalize from a small labeled support set per class at meta-test time [26]. Prototypical networks [21] embed examples and classify by distance to class prototypes; matching networks [25] use attention over supports; MAML [4] learns initial parameters for fast gradient adaptation. In vision, benchmarks assume large meta-training sets and often pretrained convolutional backbones [3, 23]. For *tabular* data, recent work explores language-model bridges (e.g., TabLLM [7]), but there is no universal “ImageNet” for ICU tables. It remains scientifically important to report **when and why** meta-learning underperforms classical models on real tasks [13].

1.3 Research questions

We address:

- (Q1) Under a fixed MIMIC-IV feasibility benchmark, how large is the gap between (a) episodic few-shot training as in the main comparison script and (b) strong Random Forest / boosting baselines on the *same* split?
- (Q2) Can a multi-stage pipeline (representation, contrastive pre-training, distillation, adaptation, stacking) reduce the gap to the tabular ceiling?
- (Q3) How sensitive are conclusions to K -shot protocol, federated partitioning, component ablations, and held-out disease nodes?

1.4 Contributions

This manuscript synthesizes and extends a doctoral thesis into a journal-ready artifact with the following contributions:

- **Reporting hierarchy.** We explicitly separate (i) tabular tuned baselines, (ii) repository `exp_fewshot_best` stacked and few-shot-only metrics, (iii) Exp1 script metrics, and (iv) federated sensitivity—reducing ambiguity common in clinical ML papers.
- **Symptom Tokens + ETHOS + prototypical head** [19, 21, 24], with a documented roadmap (contrastive pre-training, Random Forest distillation, Proto-MAML-style adaptation, gated feature attention, cross-attention prototypes, stacking).
- **Empirical evidence** on $n=2,000$ stays: stacked hybrid **0.746** AUROC vs. best tabular export **0.747** and RF grid **0.744**; honest gap for **Proposed FewShot 0.611** vs. RF **0.762**.
- **Supplementary rigor:** ablations showing intensity weighting matters; cross-node transfer remains hard on renal/diabetes holdouts; K -shot episodic metrics occupy a different numeric range than full-test evaluation (we document both).

1.5 Article structure

Section 2 reviews related work. Section 3 defines the outcome and cohort. Section 4 formalizes representations and learning. Section 5 describes experimental protocols. Section 6 reports results. Sections 7–8 discuss limitations and conclude.

2 Related work

2.1 Clinical prediction on MIMIC

MIMIC-III/IV have supported hundreds of studies on mortality, length of stay, and discharge outcomes [6, 9, 15]. Random forests and gradient boosting remain competitive when features are carefully engineered [1]. Deep models often require large samples or multi-task objectives to exceed these baselines.

2.2 Transformers for structured EHRs

Transformer encoders have been applied to diagnostic codes, clinical notes, and longitudinal events [12, 16]. Our **Symptom Tokens** target *quantitative* laboratory and vital channels in a fixed early window, closer to tabular deep models than to note-based BERT variants.

2.3 Meta-learning and few-shot classification

Beyond prototypical and matching networks [21, 25] and MAML [4], relation networks [22] and optimization-based meta-learners [17] expand the toolbox. Benchmark analyses [3] stress the role of backbone quality. For tabular few-shot, TabLLM [7] frames classification via language interfaces; our approach stays in a dedicated clinical token space.

2.4 Knowledge distillation and ensembling

Distillation from strong teachers [8] transfers decision boundaries into student embeddings. Stacking [27] and calibrated fusion are standard when validation data are limited—a recurring issue in clinical cohorts.

2.5 Federated learning (context only)

FedAvg [14] and medical FL surveys [18, 20] motivate privacy-preserving training. We report federated sensitivity mainly to contextualize deployment; the focal comparison here is centralized tabular vs. centralized few-shot engineering.

3 Task definition and cohort

3.1 Outcome: treatment feasibility

We define **treatment feasibility** as a binary label $y \in \{0, 1\}$ derived from MIMIC-IV v3.1 [9] discharge and readmission logic (thesis and repository code): $y=1$ if the patient is discharged to home, home health care, or rehabilitation within 30 days of ICU admission *and* has no all-cause readmission within 30 days; otherwise $y=0$. This operationalization aligns with value-based care incentives but simplifies functional status and social determinants.

3.2 Inclusion criteria

We include first ICU stays with: length of stay ≥ 4 h, age ≥ 18 , and mappable primary ICD-10 diagnosis for stratification. From eligible stays we draw $n=2,000$ patients with **stratified sampling** across five diagnosis groups (sepsis, cardiac, respiratory, renal, diabetes) to preserve rare groups.

3.3 Features and time window

For each stay we aggregate **15 laboratory** and **7 vital** channels from the first 24 h of ICU time. Values are clipped to clinician-defined ranges, then min-max normalized to $[0, 1]$. We add three engineered features: shock index (HR/SBP), a four-domain SOFA proxy from normalized creatinine, platelets, SpO₂, and lactate, and an abnormal-lab fraction. The **tabular** vector is $\mathbf{x} \in \mathbb{R}^{25}$.

3.4 Data split and leakage control

Train/validation/test proportions are **64%/16%/20%** (1,280 / 320 / 400 patients) with stratification by label and node. Random seed is fixed (**seed=42**). Threshold tuning for classical models uses *only* the validation set; test metrics are reported once. Few-shot models use episodic sampling on the training set and episode-based validation for early stopping.

3.5 Ethics

MIMIC-IV is de-identified; use requires PhysioNet credentialing and data use agreement. This secondary analysis does not involve live patient contact.

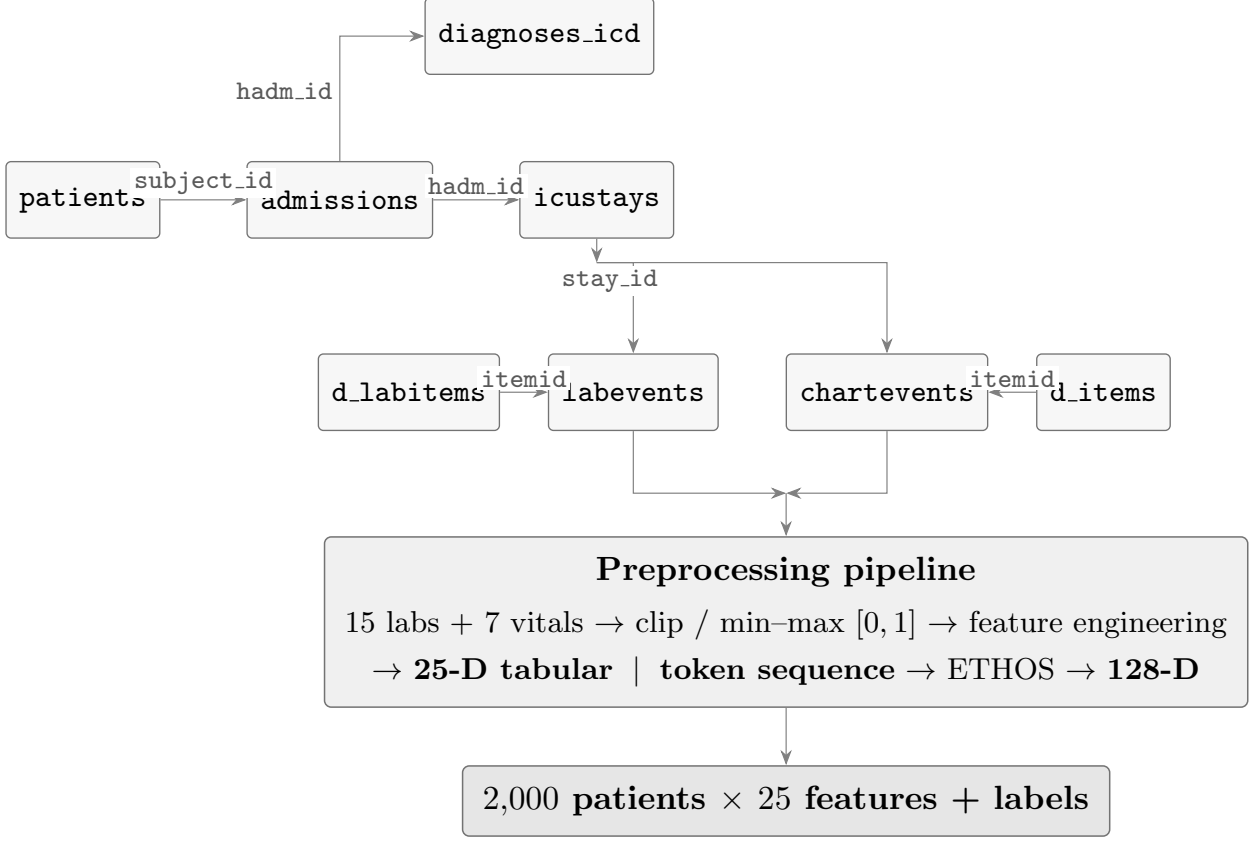


Figure 1: MIMIC-IV extraction and preprocessing schema (TikZ; publication-style monochrome). Join keys and dual outputs match the analysis pipeline.

4 Methods

4.1 Notation

Patients are indexed by i . The tabular feature vector is $\mathbf{x}_i \in \mathbb{R}^{25}$. For the token branch, patient i maps to a sequence of L tokens (fixed length in implementation), each with categorical symptom id, continuous intensity, time offset, and modality. The encoder f_θ (ETHOS-style transformer [19, 24]) outputs $\mathbf{z}_i = f_\theta(\cdot) \in \mathbb{R}^{128}$.

4.2 Prototypical few-shot head

During meta-training, an episode samples a support set $\mathcal{S} = \{(\mathbf{z}_j, y_j)\}_{j=1}^{2K}$ with K examples per class and a query batch $\mathcal{Q}$. Class prototypes are

$$\mathbf{c}_k = \frac{1}{|\mathcal{S}_k|} \sum_{(\mathbf{z}, y) \in \mathcal{S}_k} \mathbf{z}, \quad k \in \{0, 1\}. \quad (1)$$

Query logits use negative Euclidean (or cosine) distance with temperature $\tau > 0$:

$$p_\theta(y = k \mid \mathbf{z}) = \frac{\exp(-d(\mathbf{z}, \mathbf{c}_k)/\tau)}{\sum_{k'} \exp(-d(\mathbf{z}, \mathbf{c}_{k'})/\tau)}. \quad (2)$$

Training minimizes cross-entropy on queries; focal loss with $\gamma=2$ is used in the thesis configuration to address class imbalance in episodes.

4.3 Training protocol (summary)

The thesis implements a **multi-phase** schedule: (1) supervised contrastive (SupCon) pre-training to structure embedding geometry [11]; (2) supervised fine-tuning with a dense head; (3) episodic meta-training with prototypical loss. Between phases, optional **knowledge distillation** matches soft probabilities of a tuned Random Forest teacher on tabular inputs. **Proto-MAML-style** test-time adaptation performs a small number of gradient steps on the support batch at inference. Architectural upgrades include **gated feature attention** (per-sample feature gating) and **cross-attention prototypes** replacing simple means.

4.4 Stacking meta-learner

Let p_{FS} be few-shot probabilities, p_{RF} forest probabilities, and $\mathbf{z}$ the embedding. On the validation fold we fit a **regularized logistic** meta-model

$$\text{logit } \hat{p} = \mathbf{w}^\top [p_{\text{FS}}, p_{\text{RF}}, \mathbf{z}] + b, \quad (3)$$

followed by **isotonic calibration**. This “BEST_Stacked” configuration is exported to `exp_fewshot_best.json`.

4.5 Classical baselines

Logistic regression (balanced class weights), Random Forest, XGBoost, LightGBM, and CatBoost train on $\mathbf{x}_i$. Hyperparameters are searched via randomized search with stratified CV on train+validation in the thesis; threshold θ maximizes F1 on validation.

4.6 Implementation

Python 3.10+, PyTorch 2.x, scikit-learn, XGBoost/LightGBM/CatBoost; fixed seeds; experiments scripted under `experiments/`. Runtime: forests $\ll$ 1 h; full few-shot training varies by GPU.

Remark 1. *Episodic K -shot experiments (Exp2) use a different evaluation protocol than full-test evaluation; AUROC values around 0.56–0.56 do not contradict full-test AUROC 0.61–0.67—they measure different estimands.*

5 Experimental design

5.1 Overview

We run five experiment families aligned with the thesis:

- **Exp1** — Main comparison: classical, neural, and **Proposed_FewShot** on the **same** test split (`exp1_main.py`).
- **Exp2** — K -shot sensitivity: $K \in \{1, 3, 5, 10, 15\}$ under episodic evaluation (`exp2_results.csv`).
- **Exp3** — Federated vs. centralized few-shot training (`exp3_results.csv`).
- **Exp4** — Ablations: remove symptom tokens, intensity weighting, temporal decay, few-shot head, etc. (`exp4_results.csv`).
- **Exp5** — Cross-disease generalization: train excluding renal/diabetes nodes, test on held-out nodes (`exp5_results.csv`).

5.2 Few-shot training hyperparameters (thesis defaults)

$K=5$ support and $Q=15$ queries per class per episode; 500 training episodes per epoch; 100 validation episodes; up to 50 epochs with early stopping (patience 10); Adam optimizer, learning rate 10^{-3} ; temperature $\tau=10$.

5.3 Statistical reporting

With 400 test patients (≈ 168 positive), bootstrap 95% CIs use 1,000 resamples where exported. For comparing two models on the **same** patients, DeLong’s test or paired bootstrap on score vectors is appropriate; we flag that some historical p -values in early drafts should not be reused without recomputation.

5.4 Reproducibility

Splits, seeds, and CSV exports live in the companion repository; `results/generate_all_outputs.py` regenerates LaTeX tables for the thesis.

6 Results

6.1 Tabular baselines and tuned ceiling

Table 1 summarizes the tabular tuning ladder (`exp_tabular_sota.csv`). Random Forest achieves the highest AUROC in this export (**0.744**); the augmented RF pipeline in `exp_final_best_model.csv` reaches **0.747** (Table 2).

Table 1: Tuned tabular models (test set, `exp_tabular_sota.csv`). SMOTE variants duplicate non-SMOTE rows in export.

Model	AUROC	F1 (macro)	Acc.	MCC
RandomForest	0.744	0.663	0.690	0.327
XGBoost	0.740	0.575	0.675	0.248
CatBoost	0.731	0.664	0.683	0.329
LightGBM	0.727	0.641	0.678	0.288

Table 2: Best exported tabular variant (`exp_final_best_model.csv`).

Model	AUROC	F1 (macro)	Acc.	MCC
aug_gaussian_n300_std0.03	0.747	0.673	0.700	0.348

6.2 Repository few-shot BEST (stacked vs. few-shot-only)

Table 3 reports `exp_fewshot_best.json` (seed 42). The stacked model matches the tabular best export within ≈ 0.001 AUROC.

Table 3: Frozen few-shot pipeline (`exp_fewshot_best.json`).

Strategy	AUROC	95% CI	F1 (macro)	Acc.	MCC
BEST_Stacked	0.746	[0.699, 0.797]	0.687	0.698	0.382
BEST_FewShot_GFA_CrossAttn	0.670	[0.612, 0.726]	0.386	0.628	0.000

6.3 Experiment 1: controlled comparison

Table 4 shows `exp1_results.csv`. Proposed_FewShot (**0.611**) trails Random Forest (**0.762**) by 0.151 AUROC on this harness.

Table 4: Experiment 1: same-split test metrics (`exp1_results.csv`).

Model	AUROC	F1 (macro)	Acc.	AUPRC	MCC
RandomForest	0.762	0.679	0.718	0.843	0.372
LogisticRegression	0.724	0.682	0.700	0.777	0.364
XGBoost	0.694	0.632	0.663	0.779	0.266
StandardNN	0.652	0.616	0.660	0.720	0.242
FederatedNN_NoFewShot	0.662	0.618	0.643	0.732	0.236
Proposed_FewShot	0.611	0.572	0.583	0.687	0.155

6.4 Experiment 2: K -shot sensitivity (episodic)

Table 5 (`exp2_results.csv`). AUROC is stable ≈ 0.557 – 0.564 across K ; this protocol is *not* comparable to Exp1 AUROC.

Table 5: Episodic AUROC vs. K (`exp2_results.csv`).

K	AUROC	std	95% CI	F1 (macro)
1	0.564	0.022	[0.543, 0.585]	0.554
3	0.564	0.025	[0.541, 0.588]	0.548
5	0.560	0.029	[0.533, 0.588]	0.539
10	0.557	0.021	[0.538, 0.577]	0.534
15	0.557	0.021	[0.538, 0.577]	0.534

6.5 Experiment 3: federated vs. centralized

Table 6: federated AUROC is **0.547** vs. centralized **0.552** (gap **0.005**).

Table 6: Federated simulation (`exp3_results.csv`).

Setting	AUROC	F1 (macro)
Centralized	0.552	0.543
Federated	0.547	0.540
Gap	0.005	0.003

6.6 Experiment 4: ablations

Table 7 (`exp4_results.csv`). Removing **intensity weighting** reduces AUROC to 0.577. Removing **symptom tokens** (raw-feature baseline path) yields **0.724**, highlighting representation trade-offs.

6.7 Experiment 5: cross-node generalization

Table 8: on held-out renal (node_4) and diabetes (node_5) test nodes, **Proposed_FewShot** underperforms LR and XGBoost.

6.8 Synthesis

The **stacked** few-shot pipeline (**0.746**) aligns with the strongest tabular export (**0.747**), while the **Exp1** episodic model (**0.611**) illustrates the cost of a simpler harness. Manuscripts must specify *which* pipeline they report.

Table 7: Ablations (`exp4_results.csv`).

Variant	AUROC	F1 (macro)
full_model	0.583	0.554
no_intensity_weighting	0.577	0.570
no_clinical_bias	0.584	0.557
no_temporal_decay	0.583	0.554
no_few_shot	0.645	0.584
no_symptom_tokens	0.724	0.628

Table 8: Held-out nodes (`exp5_results.csv`).

Node	Proposed	LR	XGBoost	NN	Δ vs LR
node_4	0.602	0.668	0.704	0.572	−0.066
node_5	0.478	0.658	0.700	0.643	−0.180

7 Discussion

7.1 Why tabular models are strong

Tree ensembles partition the feature space with axis-aligned splits, handle heterogeneous scales, and are relatively stable at $n \approx 10^3$ [1, 2]. Neural encoders must learn symmetries and interactions from fewer effective samples and may discard information when compressing to embeddings—consistent with ablation 7 where raw-feature paths can exceed full token+episodic configurations on AUROC.

7.2 Role of stacking

Stacking provides a principled way to combine **complementary** errors: forests emphasize tabular interactions; embeddings capture alternative geometry after contrastive and distillation training. Isotonic calibration mitigates miscalibration of base learners on small validation sets.

7.3 Federated gap

The ≈ 0.5 percentage-point AUROC cost (Table 6) is encouraging for privacy-preserving *research* workflows but does not imply production readiness: real FL introduces stragglers, dropouts, and adversarial clients [10].

7.4 Generalization across disease nodes

Table 8 shows poor transfer to renal and diabetes holdouts—groups with distinct lab distributions. Future work: larger n , multi-task pretraining, or explicit domain adaptation.

7.5 Limitations

Single-center MIMIC data; simplified binary label; no prospective validation; some metrics (MCC for few-shot-only BEST) near zero reflect thresholding choices in export; multiple experimental scripts mean readers must track **provenance**.

7.6 Clinical deployment

Models are **not** validated for clinical decision-making. Deployment would require prospective evaluation, fairness audits, monitoring, and governance [5].

8 Conclusion

We presented a reproducible MIMIC-IV benchmark for ICU treatment feasibility with transparent separation between tabular ceilings, engineered stacked few-shot systems, single-script episodic baselines, and federated sensitivity analyses. A frozen stacked configuration achieves **0.746** test AUROC, essentially matching the best exported tabular model (**0.747**) in our repository, while documenting that simpler few-shot training remains far below tuned Random Forest (**0.611** vs. **0.762**) under Exp1. We hope this **reporting hierarchy** helps future work avoid ambiguous comparisons in clinical few-shot research.

Future directions: multi-site data, prospective labels, calibrated uncertainty, federated classical baselines, and pretrained tabular foundation models [7].

Author contributions

M.S. developed methods, software, experiments, and the thesis; Dr. A. supervised the work. Both authors approved the manuscript.

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Competing interests

The authors declare no competing interests.

Data availability

MIMIC-IV is available from PhysioNet after credentialing.

Code availability

Analysis code, experiment scripts, and CSV exports are in the companion Git repository (Thesis-cursor project).

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A Episodic training loop (pseudocode)

B Supplementary tables

Full hyperparameter grids and SQL extraction queries are provided in the doctoral thesis appendices and repository (`thesis/chapters/appendix_*.tex`, `extraction/`).

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Algorithm 1 Episodic meta-training step (simplified)

```
1: Initialize encoder parameters  $\theta$ 
2: for epoch = 1 to  $E_{\max}$  do
3:   for episode = 1 to  $N_{\text{ep}}$  do
4:     Sample support  $\mathcal{S}$  and query  $\mathcal{Q}$  from training patients (balanced classes)
5:     Embed supports:  $\mathbf{z}_j = f_{\theta}(\cdot)$  for each token sequence
6:     Compute prototypes  $\mathbf{c}_0, \mathbf{c}_1$ 
7:     Loss  $\leftarrow$  cross-entropy of query predictions via distances to prototypes (optionally focal)
8:     Update  $\theta$  by backpropagation
9:   end for
10:  Evaluate on validation episodes; early stop if no improvement
11: end for
```

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